Mapping Financial Behavior in Bipolar Disorder: Development of a Dataset Linking Spending Habits to Mood Fluctuations

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Introduction

Bipolar disorder is strongly associated with financial vulnerability [1]. Mood episodes in BD can correlate with fluctuations in self-awareness, impulsivity, executive functioning, and cognitive capacity. Financial decision-making during illness states is associated with a "high potential for painful consequences" [2].

Risky behaviors such as impulsive spending during hypomania [3, 4] and "comfort spending" during depressed states [5] often lead to lasting financial instability. Population-scale studies have shown individuals with BD are at a 50% greater likelihood of bankruptcy compared to the general population [6]. However, research on financial behaviors in BD is often limited to self-report.

Our prior work shows that individuals with BD are comfortable sharing financial data to inform digital financial self-management tools [7]. As such, **a primary study outcome was whether individuals would share financial data for research purposes.**

Methods

Participants (individuals with BD, N=50) from the U.S. were recruited via NIH ResearchMatch and the Depression and Bipolar Support Alliance between July 2024 and August 2025. Study sessions were conducted over Zoom. Researcher lived experience with BD was disclosed to participants at the opening of every study call.

We used Plaid's open banking data aggregation service to retrieve a two-year transaction history. We took a data minimization approach, excluding all fields containing names and locations.

Participants augmented monthly symptom recall with photographs, journals, and calendars while completing the NIMH Life Chart Method Retrospective (LCM-R). We used Qualtrics to collect the LCM-R, Monetary Choice Questionnaire 21-item, and WHODAS 2.0.

As an exploratory, hypothesis-generating analysis, we modeled symptom severity as a function of same-month financial features.

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Dataset contents

Data structure	Description
Financial data	**Two years (67,668 rows)** of dated, categorized transactions and account balances from 50 individuals
NIMH LCM Retrospective	24 concurrent months of depression/mania ratings, ordinal scale (0–3)
Participant demographics	Bipolar subtype, age, gender, race/ethnicity
Delay discounting task	21-item Monetary Choice Questionnaire
Disability assessment	World Health Organization Disability Assessment Scale 2.0

We will make this dataset available to IRB-approved researchers via the NIMH Data Archive later this year.

Descriptive case study of an individual (N-of-1)

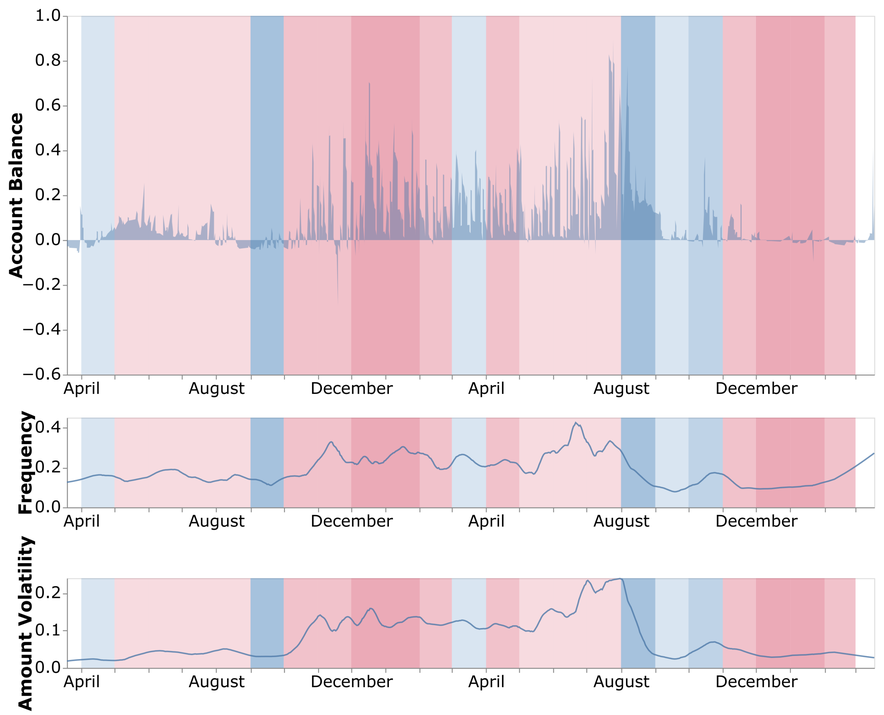


Figure 1. Normalized account balance (top), daily transaction frequency (middle), and transaction amount volatility (bottom, 14-day rolling standard deviation) for a single individual. Mood symptoms (mania in red, depression in blue) reflect NIMH LCM-R ratings with **darker shading representing increased mood severity**. For this participant, transaction frequency and volatility show a marked increase during hypomania.

Exploratory analysis

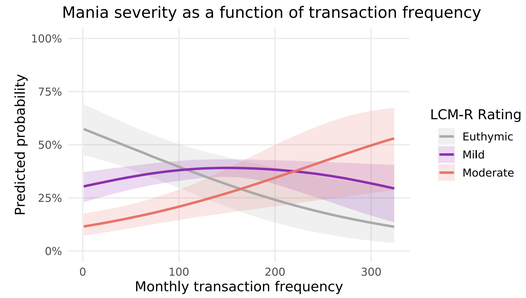


Figure 2. Months with greater transaction frequency were credibly associated with higher mania severity (OR ~1.43 per SD, 95% CI 1.15-1.87). Lines show posterior median probabilities for severity levels; shaded ribbons denote 95% credible intervals. We excluded "severe" rating in this visualization as it was <3% of our sample.

Discussion

We demonstrate the feasibility of collecting longitudinal financial data and retrospective symptom trajectories from individuals with BD.

Transaction frequency appears to be associated with increased mania severity. Future confirmatory work should assess bidirectional lead-lag relationships via temporal network analysis.

Prospective studies should include exogenous measures such as EMA, transaction tagging, diaries, and life event reporting to contextualize money behaviors and assess potential bidirectional dynamics of money, life circumstances, and mental health.

Interventions (digital or clinical) can be informed by validated psychological theory (e.g., DBT skills, ACT) as well as objective money behaviors.

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